

Financial Inclusion & Digital Lending Business Intelligence Project

Project Brief & Business Requirements Document

Client (simulated)	Wekeza Analytics — a fictional Kenyan fintech intelligence consultancy
Engagement type	Business Intelligence Dashboard & Portfolio Risk Analytics
Market	Kenya — mobile money, agency banking & digital micro-lending
Dataset provided	kenya_fintech.db (SQLite, 7 relational tables, ~26,600 rows)
Prepared for	Data Analyst / BI Developer candidate portfolio project

1. Project Scenario

Kenya's financial services market is defined by mobile money (M-Pesa, Airtel Money), agency banking, and a fast-growing digital micro-lending sector (M-Shwari, KCB M-Pesa, Fuliza, Tala, Branch, Zenka, the government-backed Hustler Fund, and bank-led products like Timiza). This mix drives huge financial inclusion gains but also real business risk: agent liquidity gaps, uneven rural/urban adoption, and loan portfolios where default and late repayment quietly erode margins.

Wekeza Analytics (a fictional consultancy created for this portfolio project) has been engaged by a mid-sized digital lender and mobile money aggregator to build a Business Intelligence solution that gives leadership a single view of customer growth, transaction behaviour, agent network health, and loan portfolio risk across Kenya's 47 counties.

2. Your Role

You are the BI Analyst on this engagement. You have been handed the raw relational database (`kenya_fintech.db`) extracted from the client's transaction and lending systems. Leadership does not want raw tables — they want a decision-ready dashboard and a short set of insights they can act on this quarter.

3. Data Provided

The database `kenya_fintech.db` is a normalized relational model with the following tables. Import it into Power BI, Tableau, Metabase, or query it directly with SQL / pandas.

Table	Rows	Key Fields	Purpose
counties	30	county_id, region, population	Geographic dimension (Kenyan counties)
customers	1,600	customer_id, segment, kyc_tier, age, gender	Customer dimension
agents	220	agent_id, agent_type, county_id, tier	Mobile money / bank agent network
loan_products	8	product_id, provider, interest_rate, tenure	Digital lending product catalogue
transactions	18,000	transaction_id, type, channel, amount, time	Mobile money transaction fact table
loans	3,400	loan_id, principal, interest, status	Loan disbursement fact table
repayments	3,145	repayment_id, loan_id, days_late	Repayment fact table (links to loans)

Relationships: customers → county (many-to-one) · transactions → customer & agent · loans → customer & loan_product · repayments → loan (one-to-many).

4. Business Questions the Dashboard Must Answer

Leadership has grouped their questions into four themes. Your dashboard should let a non-technical executive answer every question below in under 3 clicks.

A. Financial Inclusion & Customer Growth

- How is customer registration trending, and which counties are under-penetrated relative to their population?
- What share of customers are on Tier 1 vs Tier 3 KYC, and how does that vary by county classification (Urban/Mixed/Rural)?

- How does smartphone access differ by customer segment (e.g. Smallholder Farmers vs Salaried), and what does that imply for app vs USSD channel strategy?
- Which customer segments are growing fastest and should be prioritised for the next acquisition campaign?

B. Transaction & Channel Performance

- What is the monthly transaction volume and value trend across 2024–2025, and are there seasonal spikes (e.g. school fees, festive season)?
- Which transaction types (Send Money, Pay Bill, Buy Goods, Withdraw, Deposit, Airtime, Remittance) generate the most fee revenue?
- How does channel usage (USSD vs App vs Agent-Assisted) differ between rural and urban counties?
- Which agents / counties have the highest transaction volume, and where is the agent network under-serving demand?

C. Loan Portfolio & Risk

- What is the current loan book split by status: Repaid, Repaid Late, Active, Defaulted?
- Which loan products (M-Shwari, Tala, Branch, Fuliza, Hustler Fund, etc.) carry the highest default rate, and is that linked to interest rate or tenure?
- What is the Portfolio at Risk — total outstanding principal tied to Defaulted and Active loans?
- How does average days-late-on-repayment vary by customer segment and county — where should credit scoring be tightened?

D. Executive Summary

- If leadership had 60 seconds, what are the 3 headline KPIs and the 3 biggest risks or opportunities?
- What one recommendation would you make for Q3 based on the data (e.g. expand agent network in county X, tighten credit for product Y)?

5. Suggested KPIs (build these, then decide if the client needs more)

KPI	Definition
Total Active Customers	COUNT(customers)
Transaction Value (KES)	SUM(transactions.amount) — filterable by month/type/channel
Fee Revenue (KES)	SUM(transactions.transaction_fee)
Loan Disbursement Value	SUM(loans.principal_amount)
Portfolio at Risk (PAR)	SUM(principal_amount) WHERE status IN ('Defaulted','Active') ÷ total disbursed
Default Rate	COUNT(loans WHERE status='Defaulted') ÷ COUNT(loans)
Average Days Late	AVG(repayments.days_late) WHERE days_late > 0
Agent Density	COUNT(agents) per 100,000 population, by county
Smartphone Penetration	AVG(is_smartphone_user) by segment / county

6. Deliverables Expected

- **SQL layer:** a set of clean, commented SQL queries (views or CTEs) answering each business question in Section 4.
- **Dashboard:** built in Power BI, Tableau, or Looker Studio — one Executive Overview page plus 2–3 drill-down pages (Growth, Transactions, Loan Risk).
- **County-level choropleth map** showing at least one metric (e.g. transaction value or default rate) across Kenya.
- **Write-up (1 page):** three headline insights and one data-backed recommendation, written for a non-technical executive.

7. Why This Project Stands Out in a Portfolio

Most student BI portfolios reuse the same public Superstore or Netflix datasets. This project uses a locally-relevant, self-built relational database modelled on Kenya's real fintech ecosystem — mobile money, agency banking, and digital lending — which signals domain awareness to Kenyan employers (banks, telcos, fintechs, and consultancies) and gives you a genuine SQL + data modelling story to tell in interviews, not just a dashboard screenshot.

Note: All data in `kenya_fintech.db` is synthetically generated for portfolio/demo purposes. It does not represent real customers, agents, or institutions.